

Homework 8

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Our study focuses specifically on Crude Oil Prices as a high-frequency economic indicator. Unlike quarterly GDP or monthly Unemployment reports, oil prices fluctuate daily, providing a real-time ‘stress test’ for urban environments like Dallas.

```
# Loading all the required libararies
library(ggplot2) # For plotting
library(dplyr) # For signifigance test
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(lubridate) # For making program able to read dates from Dallas_Crimes dataset
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
# --- 2. LOAD & CLEAN OIL DATA ---
oil_raw <- read.csv("Crude_Oil_Prices.csv")

# Cleans dataset to make everything more readable
oil_price_data <- oil_raw %>%
  mutate(date_clean = as.Date(observation_date),
         price = as.numeric(DCOILWTICO)) %>%
  filter(date_clean >= "2020-01-01") %>%
  filter(!is.na(price))

# Calculate the Median ($71.68)
oil_median_value <- median(oil_price_data$price, na.rm = TRUE)
```

```
# --- 3. LOAD & CLEAN DALLAS CRIME DATA ---
dallas_raw <- read.csv("Dallas_Crimes_dallas_Website.csv")
```

```
# Turn 852k rows into a Daily Count
daily_summary <- dallas_raw %>%
  mutate(date_clean = as.Date(Date1.of.Occurrence)) %>%
  group_by(date_clean) %>%
  summarize(crime_volume = n()) %>%
  filter(!is.na(date_clean))

head(daily_summary)
```

```
## # A tibble: 6 x 2
##   date_clean crime_volume
##   <date>         <int>
## 1 1982-02-01           1
## 2 1984-08-21           1
## 3 1989-07-12           1
## 4 1992-09-15           1
## 5 1993-01-03           1
## 6 1995-06-01           1
```

```
# --- 4. THE JOIN (The Handshake) ---
master_analysis <- daily_summary %>%
  inner_join(oil_price_data, by = "date_clean")
```

```
# --- 5. THE GROUPING (Creating the 2 Levels) ---
master_analysis <- master_analysis %>%
  mutate(oil_status = ifelse(price > oil_median_value, "High_Oil", "Low_Oil")) %>%
  filter(!is.na(oil_status)) %>%
  mutate(oil_status = as.factor(oil_status))

head(master_analysis)
```

```
## # A tibble: 6 x 6
##   date_clean crime_volume observation_date DCOILWTICO price oil_status
##   <date>         <int> <chr>             <dbl> <dbl> <fct>
## 1 2020-01-02       344 2020-01-02         61.2  61.2 Low_Oil
## 2 2020-01-03       373 2020-01-03         63    63  Low_Oil
## 3 2020-01-06       392 2020-01-06         63.3  63.3 Low_Oil
## 4 2020-01-07       366 2020-01-07         62.7  62.7 Low_Oil
## 5 2020-01-08       327 2020-01-08         59.6  59.6 Low_Oil
## 6 2020-01-09       372 2020-01-09         59.6  59.6 Low_Oil
```

```
# --- 6. THE GRAND FINALE: T-TEST ---
# This proves if High Oil days have significantly different crime than Low Oil days
t_result <- t.test(crime_volume ~ oil_status, data = master_analysis)
```

```
# --- 7. THE RESULTS FOR YOUR SLIDES ---
print(paste("The Median Oil Price is:", oil_median_value))
```

```
## [1] "The Median Oil Price is: 71.68"
```

```
print(t_result)
```

```
##  
## Welch Two Sample t-test  
##  
## data: crime_volume by oil_status  
## t = 5.5711, df = 1537.3, p-value = 2.984e-08  
## alternative hypothesis: true difference in means between group High_Oil and group Low_Oil is not equal to 0  
## 95 percent confidence interval:  
## 8.087625 16.877545  
## sample estimates:  
## mean in group High_Oil mean in group Low_Oil  
## 375.4493 362.9667
```

Based on our two sided t-test, we got a p-value of 2.984e-08, which is much less than 0.05. This is statistically significant and this means we have to reject the null hypothesis which was the oil and dallas crimes have no correlation

There are approximately 13 more crimes committed per day during times of high oil prices compared to low oil prices, which suggests there is a strong positive correlation between oil prices and the amount of crime committed in the city of dallas

Graph

```
# 1. THE PLOT CODE (No ggthemes needed)  
ggplot(master_analysis, aes(x = oil_status, y = crime_volume, fill = oil_status)) +  
  # The Violin: Shows the "shape" and density of 852k reports  
  geom_violin(alpha = 0.5, color = "white") +  
  # The Boxplot: Provides the formal median and quartiles  
  geom_boxplot(width = 0.1, color = "black", outlier.shape = 16, outlier.size = 1, outlier.alpha = 0.1)  
  # The Colors: Using a bold Red vs. Professional Blue  
  scale_fill_manual(values = c("High_Oil" = "#E31A1C", "Low_Oil" = "#1F78B4")) +  
  # The Storytelling Labels  
  labs(  
    title = "Energy Costs vs. Public Safety: The Dallas Correlation",  
    subtitle = "Welch Two-Sample T-Test | t-score = 5.57 | p < 0.0000001",  
    x = "Oil Price Status (Split at Median: $71.68)",  
    y = "Daily Reported Incidents",  
    caption = "Data: Dallas Open Data (N=852,447) & WTI Crude Spot Prices"  
  ) +  
  # 2. THE CUSTOM THEME (Making it look "Blow People Away" manually)  
  theme_minimal(base_size = 14) +  
  theme(  
    legend.position = "none",
```

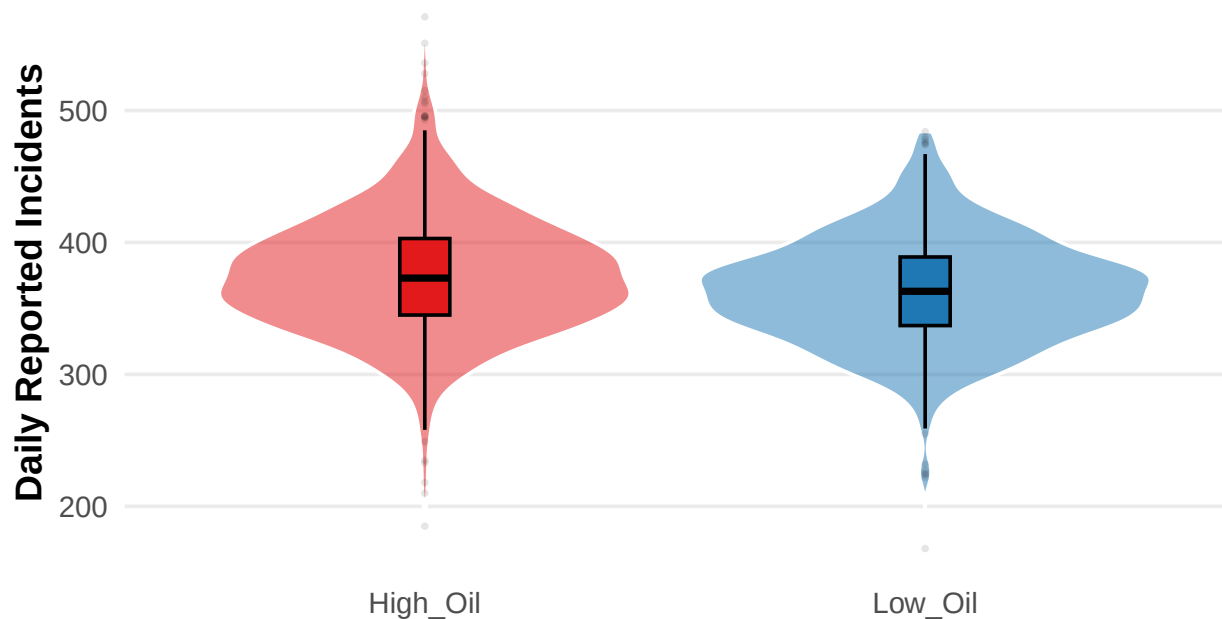
```

plot.title = element_text(face = "bold", size = 16, color = "#222222"),
plot.subtitle = element_text(size = 12, color = "#555555"),
axis.title = element_text(face = "bold"),
panel.grid.minor = element_blank(),
panel.grid.major.x = element_blank() # Cleans up the vertical lines
)

```

Energy Costs vs. Public Safety: The Dallas Correlation

Welch Two-Sample T-Test | t-score = 5.57 | $p < 0.0000001$



Oil Price Status (Split at Median: \$71.68)

Data: Dallas Open Data (N=852,447) & WTI Crude Spot Prices

we are using a Welch Two-Sample T_Test, because it doesn't assume equal variance between the two groups

the reason we are using the boxplot is to show the median, IQR, range, and the outliers

the reason we are using the violin plot is to show the distribution (Probability density) which is the width of the bell. The main thing to note here is even though they both are bell-shaped which shows they have a relatively normal distribution, the fatness of the violin shows that the probability of a higher number of crimes occurring (around 373) is now much more likely than it was when oil prices were low.

Future Plans: Rather than just using oil, we would like to use other variables to determine whether crime occurs more during an economically bad period. Although oil prices is one of the main factors, alone it is not enough to conclude how bad the economic period really is. We would need more data such as inflation rates and unemployment data. We could also use a linear regression model to predict how much more crime could be committed based on how much the previous data increased.